Scientific Computing Exercise Set 3

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***Repository**: https://github.com/XthPB/SCI_COMP_set3*

I. Introduction

THIS final assignment presents two open challenges. The first challenge is about the incompressible Navier-Stokes equations. We simulate flow past a cylinder using three numerical methods: finite difference (FD), finite element (FEM via NGSolve [1]), and lattice Boltzmann (LBM). We compare their accuracy, stability limts, and computational cost. The second challenge applies the Helmholtz equation to optimise WiFi router placement in a complex indoor enviroment.

II. Theory

A. Navier-Stokes Equations

The incompressible Navier–Stokes equations for a fluid with constant density ρ and kinematic viscosity ν are

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u}\cdot\nabla)\mathbf{u} = -\frac{1}{\rho}\nabla p + \nu\nabla^2\mathbf{u}, \quad \nabla\cdot\mathbf{u} = 0, \quad (1)$$

where $\mathbf{u}$ is the velocity vector and p is the pressure.

The Reynolds number $\mathrm{Re} = UD/\nu$ is an important parameter. U is the mean inflow velocity and D is cylinder diameter. When Re is low the flow is steady and symmetric. Above a critical value ($\mathrm{Re} \approx 47$ for unbounded flow), the wake becomes unsteady and vortex shedding starts. The shedding frequency f is given by the Strouhal number $\mathrm{St} = fD/U$.

B. Problem Setup and Boundary Conditions

The domain follows the DFG benchmark geometry described in [2]: a rectangular channel of length $L = 2.2\,\mathrm{m}$ and height $H = 0.41\,\mathrm{m}$, with a circular cylinder of diameter $D = 0.1\,\mathrm{m}$ centred at $(0.2, 0.2)\,\mathrm{m}$. In nondimensional units with $D = 1$ and $U_{\text{mean}} = 1$, this gives $L/D = 22$, $H/D = 4.1$, cylinder centre at $(2, 2)$, and radius 0.5. The blockage ratio $D/H = 0.244$ is intentionally high, leading to elevated Strouhal numbers compared to unbounded flow.

The boundary conditions are as follows. At the inlet, a parabolic velocity profile is prescribed:

$$u(y) = \frac{4U_m\, y\,(H-y)}{H^2}, \quad v = 0, \quad (2)$$

with $U_m = 1.5$, giving a mean inflow velocity of approximately 1. At the top and bottom walls, no-slip conditions are applied ($\mathbf{u} = 0$). At the outlet, a convective (Sommerfeld) boundary condition $\partial_t\mathbf{u} + U_c\,\partial_x\mathbf{u} = 0$ is used to suppress spurious pressure-wave reflections. On the cylinder surface, no-slip conditions are enforced.

1) Finite Difference: Chorin Projection Method: THe Chorin projection method [3] splits the time steps into three parts: (i) advance a tentative velocity $\mathbf{u}^*$ using advection and diffusion without the pressure gradient; (ii) solve the pressure Poisson equation $\nabla^2\phi = \nabla\cdot\mathbf{u}^*/\Delta t$; and (iii) correct the velocity via $\mathbf{u}^{n+1} = \mathbf{u}^* - \Delta t\,\nabla\phi$ to enforce $\nabla\cdot\mathbf{u} = 0$.

To avoid checkerboard decoupling while retaining the compact 5-point Laplacian stencil, the variables are stored on a collocated grid, with half-cell (one-sided) differences (backward for divergence and forward for gradient). The cylinder is handled via volume penalisation [4], adding an implicit Brinkman drag $-\chi\mathbf{u}/\eta$ (mask χ, parameter η) that avoids the explicit restriction $\Delta t \le \eta/2$. Advection uses a second-order linear upwind scheme.

2) Finite Element: Taylor-Hood with IMEX Euler: An unstructured triangular mesh from Netgen/OCC [1] models the cylinder geometry exactly, removing the staircase errors present in structured grids. Velocity and pressure are discretised with Taylor–Hood P_2/P_1 elements [5], satisfying the inf-sup (LBB) condition and avoiding spurious pressure modes.

Time integration follows an IMEX Euler scheme. In this scheme diffusion and pressure are treated implicitly. The nonlinear convection term is evaluated explicitly from the previous time step. Drag and lift are computed by integrating the boundary stress $(-p\mathbf{I} + \nu\nabla\mathbf{u})\cdot\mathbf{n}$ over the cylinder surface.

3) Lattice Boltzmann: D2Q9 with MRT Collision: The D2Q9 lattice Boltzmann method evolves nine distribution functions f_i on a Cartesian lattice, recovering macroscopic density and velocity as moments. The BGK collision model [6] relaxes all f_i toward equilibrium at a single rate $1/\tau$, where $\tau = 0.5 + \nu_{\text{lat}}/c_s^2$; as Re increases

at fixed resolution $\tau \to 0.5$ and non-hydrodynamic ghost modes oscillate rather than decay, causing instability.

We therefore use the MRT operator [7], which transforms f_i into moment space via $\mathbf{M}$ and relaxes each mode independently: conserved moments are unchanged, stress modes relax at $s_\nu = 1/\tau$ (preserving the physical viscosity), and ghost modes at $s_{\text{ghost}} = 1.1$, decoupling viscosity from non-physical-mode damping and substantially improving stability near $\tau = 0.5$. Inlet and outlet boundary conditions follow Zou–He [8]; no-slip walls and the cylinder use bounce-back.

C. WiFi signal model: Helmholtz approximation

The propagation of WiFi signals in an indoor environment can be approximated using wave theory. While the full electromagnetic behavior is described by Maxwell's equations, a simplified scalar approximation is sufficient to capture the dominant effects relevant for signal coverage. This leads to the Helmholtz equation:

$$\nabla^2 u + k^2 u = f(x, y), \tag{3}$$

where $u(x, y)$ denotes the complex wave field, k is the wavenumber, and $f(x, y)$ represents the signal source.

In indoor environments with walls and obstacles, the propagation properties vary spatially. This effect is incorporated through a position-dependent effective wavenumber:

$$k_{\text{local}} = n(x, y)\, k, \tag{4}$$

where $n(x, y)$ is a complex refractive index. Its real part modifies the phase velocity of the wave, while its imaginary part introduces attenuation, modeling energy loss in materials such as walls.

The source is modeled as a localized Gaussian:

$$f(x, y) = A \exp\left(-\frac{(x - x_r)^2 + (y - y_r)^2}{2\sigma^2}\right), \tag{5}$$

Here, we set $A = 10^4$ and $\sigma = 0.2\,\text{m}$ for the source and assume a uniform wall thickness of $0.15\,\text{m}$ throughout the environment. (x_r, y_r) is the router position. This provides a smooth approximation of a point emitter and ensures numerical stability in the discretized system.

Since signal strength is usually measured on a logarithmic scale, the field magnitude is converted to decibel units:

$$u_{\text{dB}}(x, y) = 20 \log_{10}(|u(x, y)|), \tag{6}$$

To obtain a physically meaningful estimate of signal strength, the field is averaged over a small disc shaped region around each measurement location. The overall performance of a router placement is defined as the sum of signal strength across all measurement locations. Taking the average in dB scale corresponds to taking a geometric mean in linear amplitude, which tracks with how signal strength is reported in practical wireless systems.

III. Methods

A. Implementation

All simulations were implemented in **Python**. Numerical operations were carried out using **NumPy**, and visualisation and animation used **Matplotlib**. **Numba** was used to accelerate performance-critical loops. The finite element solver employed **NGSolve** with Netgen/OCC for exact geometry meshing. Sparse linear solves and LU factorisations for the Helmholtz equation were performed with **SciPy**, while FFT-based convolutions accelerated objective evaluations.

B. FD Implementation Details

The Poisson system is assembled once as a sparse matrix and factored via `scipy.sparse.linalg.splu`, with the LU decomposition reused at every time step. The time step is chosen adaptively as the minimum of the advective CFL limit $\Delta t_{\text{adv}} = C/(|u|/\Delta x + |v|/\Delta y)$ ($C = 0.25$), the diffusive limit $\Delta t_{\text{diff}} = \Delta x^2 \Delta y^2 / [4\nu(\Delta x^2 + \Delta y^2)]$, and a hard cap of 0.05. The flow is initialised with the parabolic inflow profile projected to be divergence-free, with a small random perturbation added in the near-wake to trigger shedding.

C. FEM Implementation Details

Using NGSolve [1], the combined mass-stiffness-pressure system $\mathbf{A} = \frac{1}{\Delta t}\mathbf{M} + \nu\mathbf{K} + \mathbf{B}$ is assembled once and factored via UMFPACK. At each step the explicit convection form $-(\nabla \mathbf{u} \cdot \mathbf{u}^n, \mathbf{v})$ is reassembled, the right-hand side is formed, the Dirichlet inlet condition is lifted, and the solution is updated with the pre-factored inverse. Mesh refinement near the cylinder is controlled via the `maxh` parameter.

D. LBM Implementation Details

Physical parameters map to lattice units via D_{lat} (cells per diameter), U_{lat} (lattice velocity, setting Ma =

U_{lat}/c_s), and $\tau = 0.5 + U_{\text{lat}} D_{\text{lat}}/(\text{Re} \cdot c_s^2)$, giving grid dimensions $N_x = \lfloor (L/D)\, D_{\text{lat}} \rfloor + 1$, $N_y = \lfloor (H/D)\, D_{\text{lat}} \rfloor + 1$. Streaming uses direct slice assignment (roughly $2\times$ faster than `np.roll`), and the MRT collision kernel is JIT-compiled with Numba [9] using `prange` for row-level parallelism. Cylinder forces are obtained via the momentum-exchange method.

E. Validation and Stress-Testing Strategy

We validate all three methods at $\text{Re} = 20$ (steady symmetric wake with no shedding) and $\text{Re} = 100$ (periodic vortex shedding). The Strouhal number is extracted from the lift signal via FFT after discarding the first 30–50% of the time series as initial transient. We compare against the DFG blocked-channel reference [10], which reports $\text{St} = 0.295$–0.305 for this geometry.

To find the stability limits, we apply a sweep on the Reynolds number from $\text{Re} = 20$ to $\text{Re} = 500$. Moderate resolutions were used for each methods. To assess grid convergence we also ran each method at resolution levels at $\text{Re} = 100$.

F. WiFi router position optimization

The Helmholtz equation was solved on a 2D uniform grid using a finite difference discretisation of the Laplacian with the standard five-point stencil. The resulting sparse linear system was solved using a direct LU factorization to be reused throughout the optimization pipeline.

To reduce computational cost, a scaled wavenumber $k = 1/3$ was used, corresponding to an effective frequency of approximately 0.8 GHz ($\lambda_{\text{eff}} \approx 38$ cm). This increases the wavelength relative to the grid resolution, reducing numerical stiffness and allowing coarser discretization without loosing the qualitative behavior of the WiFi signal.

The house layout was encoded as a material grid, where each cell is assigned a complex refractive index representing either free space or walls. Absorbing boundary conditions were applied, to reduce unwanted reflections at the domain boundaries.

Signal strength was evaluated in decibel scale and averaged over a small circular region (radius 5 cm) around each of the measurement locations. Measurement regions are encoded by the indicator matrix E, where each column corresponds to a spatial averaging mask centered at the measurement location. The objective function is then defined as the sum of these averaged values which is to be maximized.

A baseline hierarchical grid search with Nelder-Mead refinement was used as a reference solution. To improve efficiency, we introduce an adjoint-based method that allows simultaneous evaluation of all potential router positions and enables gradient-based refinement.

Algorithm 1 WiFi router optimisation (adjoint-based)

```
Assemble system matrix A and compute LU factorisation
Construct measurement indicator matrix E
Compute adjoint fields using Algorithm 2
Evaluate objective over all candidate positions using Algorithm 3
Select top candidate positions
for each selected candidate do
    Compute gradient using Algorithm 4
    Refine position using L-BFGS-B
end for
Return optimal router position
```

Algorithm 2 Adjoint field computation

```
Solve multi right-hand side system:
A^T W = E
Each column w_k corresponds to adjoint field at measurement node m_k
```

Algorithm 3 FFT-based evaluation of source response

```
Precompute FFT of Gaussian kernel g
for each adjoint field w_k do
    ŵ_k = F(w_k)
    û_k = ŵ_k · ĝ
    u_k = F^{-1}(û_k)
end for
Evaluate objective at all feasible router positions
```

Algorithm 4 Gradient evaluation for router position

```
for each parameter θ ∈ {x_r, y_r} do
    ∂f_i/∂θ = f_i · (x_i − θ)/σ^2
    Solve:
    A ∂u/∂θ = ∂f/∂θ
    ∂S/∂θ = Σ_m (1/|u|) ∂u/∂θ
end for
```

IV. Results

A. Steady Wake Validation at Re = 20

All three methods produce the expected wake (a steady, symmetric recirculation zone behind the cylinder with

no vortex shedding). The velocity fields show a smooth wake closing approximately 2-3 diameters downstream of the cylinder. This agreement across all three methods confirms that the basic boundary conditions and flow initialization are correctly implemented for each solver.

B. Unsteady Validation at Re = 100

At Re $= 100$, the flow for LBM and FEM show clear periodic vortex shedding. Fig. 1 displays the vorticity fields for all three methods. The FD solution shows a more diffused wake without clear periodic shedding. This tells us that the volume penalization boundary treatment introduces excessive numerical diffusion that suppresses the physical instability.

Table I compares the extracted Strouhal numbers against the DFG reference [10]. FEM matches the benchmark precisely with St $= 0.300$, while LBM-MRT is within 1.2% at St $= 0.304$, consistent with the half-cell shift in the effective wall position introduced by bounce-back boundaries. The extracted Strouhal number for the FD method is St $= 0.015$. This indicates no periodic shedding. This is due to the combination of volume penalization [4] and the staircase cylinder geometry. This smears the boundary layer and causes numerical diffusion that dampens the wake instability.

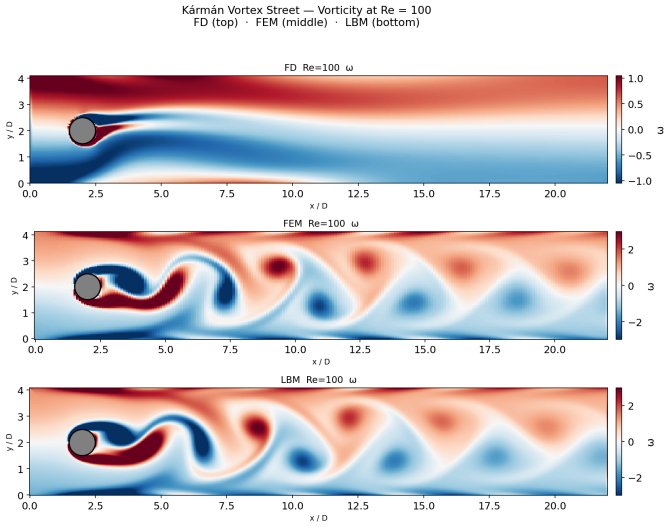

Fig. 1: Vorticity field $\omega = \partial v/\partial x - \partial u/\partial y$ at Re $= 100$ for FD (top), FEM (middle), and LBM (bottom). Red and blue denote positive and negative vorticity, respectively.

TABLE I: Strouhal number comparison at Re $= 100$. The DFG benchmark reference [10] for this blocked channel ($D/H = 0.244$) is St $= 0.295$–0.305.

Method	St	Deviation from DFG
FD (660×123)	0.015	no shedding
FEM (maxh = 0.06)	0.300	+0.0%
LBM-MRT ($D_{\text{lat}} = 50$)	0.304	+1.2%

C. Reynolds Number Sweep and Stability Limits

We sweep Re $= 20$–500 at moderate resolution. Results are shown in Table II. The FD method stays stable beyond Re $= 500$, though this reflects numerical diffusion suppressing the wake instability rather than physical accuracy. LBM-MRT at $D_{\text{lat}} = 50$ fails at Re $= 500$ as τ approaches the hard limit of 0.5, the maximum value at which LBM-MRT was found to be stable was Re $= 450$. FEM becomes unstable at Re $= 200$ due to the explicit treatment of the convective term. Above Re ≈ 189 the wake is physically three-dimensional [11], so all higher-Re results are 2-D only.

TABLE II: Stability limits from the Re sweep.

Method	First failure	Note
FD (220×41)	> 500	num. diffusion
LBM-MRT ($D_{\text{lat}} = 50$)	500	$\tau \to 0.5$
FEM (maxh = 0.15)	200	explicit CFL

D. Resolution Study at Re = 100

To assess grid convergence, we run each method at two resolution levels at Re $= 100$. Doubling the FD grid from 200×38 to 440×82 shifts the Strouhal number from 0.216 to 0.239, yet it still does not approach the benchmark value. This indicates that the dominant error is not the discretisation error but rather the $O(\sqrt{\eta})$ smearing of the no-slip boundary introduced by volume penalisation, which persists regardless of grid refinement. For LBM, doubling D_{lat} from 30 to 60 preserves St at approximately 0.310 while the drag coefficient decreases from 9.0 to 6.4, showing grid convergence in the force computation toward physically reasonable values. FEM with maxh refined from 0.25 to 0.12 gives St $= 0.286$ and a mean $C_D = 2.28$, close to the expected value from the DFG benchmark [10], confirming second-order convergence of the Taylor–Hood elements.

E. Computational Cost and Method Comparison

Table III summarizes the key trade-offs at Re $= 100$. FEM's wall time of $\sim$38,700 s dwarfs the others because UMFPACK refactors a global sparse system at every step. LBM's purely local collision–stream kernel is the natural fit for GPU acceleration, while FD falls between the two since only its Poisson solve requires global communication.

TABLE III: Method comparison summary at Re = 100.

Property	FD	FEM	LBM
Typical DOFs	∼9k	∼41k	∼226k
Wall time [s]	182	37 000	1 744
Solver bottleneck	Poisson LU	UMFPACK	local ops
Curved geometry	staircase	exact	staircase
GPU suitability	moderate	poor	excellent
Implementation effort	∼150 LOC	∼80 LOC	∼120 LOC
Max stable Re	>500	200	450

F. Objective landscape and candidate selection

The adjoint formulation evaluates the objective over all feasible router positions in a single iteration, producing the score landscape shown in Figure 2. High values concentrate near the central corridor and door openings, where signal propagation encounters fewer obstructions, while positions enclosed within rooms yield lower scores due to wall attenuation. This global pool was then used to select a small set of promising candidates for further local optimization.

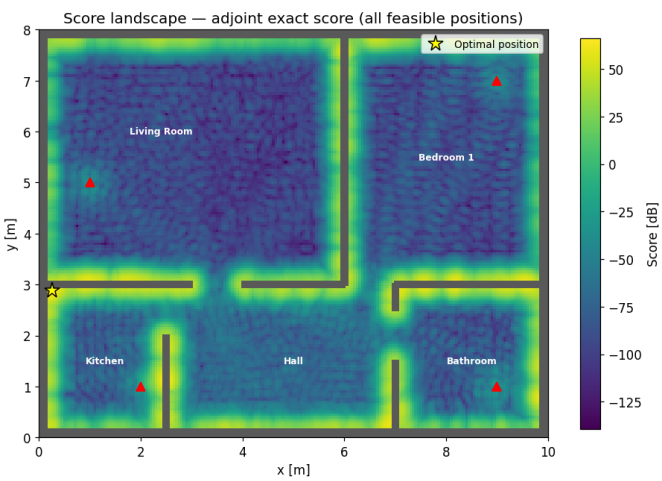

Fig. 2: Adjoint-based objective landscape over feasible router positions.

G. Optimal placement and resulting field

Local optimisation converges to an optimal position at (0.25, 2.89) m, located near the top kitchen wall, which provides a balance between coverage reach and attenuation. The resulting signal field (Figure 3) shows stronger coverage in the kitchen and living room, with decreased, yet non-zero levels in the bedroom and bathroom, indicating modest coverage across all measurement points.

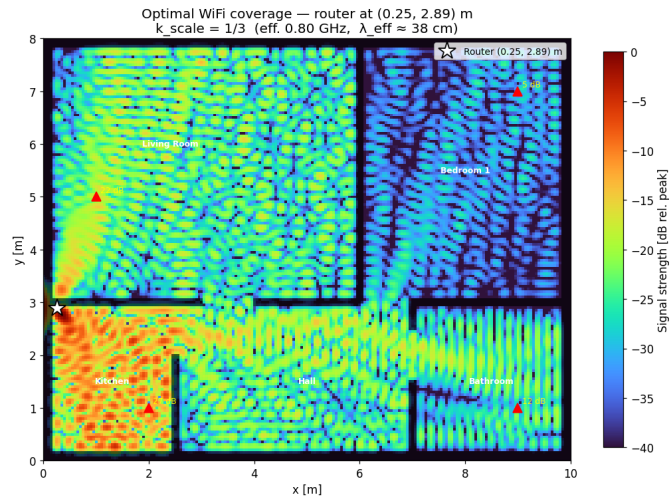

Fig. 3: Signal field for the optimal router placement.

H. Room-wise signal distribution

The average signal strength per room as summarised in Figure 4 : Living room (21.8 dB), kitchen (27.1 dB), bathroom (12.4 dB) and bedroom (4.9 dB). The effects of distance and wall attenuation are well reflected in these results, with open regions closer to the router achieving higher values.

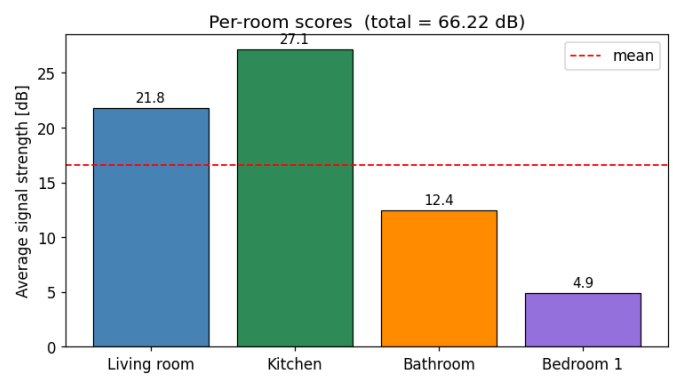

Fig. 4: Average signal strength per room.

I. Objective value and computational performance

The final objective value acheived was:

$$S = \sum_k S_k = 66.22 \text{ dB},$$

which serves as a coverage metric all across rooms. The adjoint approach reaches the same optimum router placement as the baseline method at significantly lower computational cost.

TABLE IV: Baseline vs adjoint-based optimisation

Component	Baseline	Adjoint
Global search	∼3400 evals	∼27k scan
Linear solves	Per candidate	Single solve
Evaluation	Full solve	FFT convolution
Local refinement	∼300 evals	∼40 evals
Runtime	∼14 s	∼1.4 s
Speedup	∼10×	

V. Discussion

A. Comparing Numerical Solvers

The FD projection method [3] is the most numerically robust, surviving beyond Re $= 500$. However, this robustness is misleading: the staircase cylinder and volume penalization [4] smear the no-slip condition by $O(\sqrt{\eta})$, damping the physical instabilities that drive shedding. Even at Re $= 100$ the solver fails to produce a Kármán street. Doubling the grid shifts St from 0.216 to 0.239 but the penalization error dominates, so a body-fitted mesh or sharper immersed boundary would be needed to resolve shedding correctly. This illustrates that numerical stability does not imply physical accuracy.

FEM [5], [1] is the most accurate method. Exact cylinder geometry via Netgen/OCC eliminates staircase error, and the inf-sup stable P_2/P_1 pair delivers St $= 0.300$, matching the DFG benchmark [10] to within rounding. The main drawback is cost: the UMFPACK direct solve dominates the 38,700 s wall time and is serial, making the method difficult to parallelize without switching to iterative solvers.

LBM-MRT [7] balances accuracy and efficiency. Despite staircase bounce-back boundaries (effective wall shifted by $\Delta x/2$), it recovers St $= 0.304$. We chose MRT over BGK [6] because BGK relaxes all moments at $1/\tau$, causing ghost mode oscillations as $\tau \to 0.5$. MRT damps these independently at $s_{\text{ghost}} = 1.1$, extending the stability limit to Re $= 400$ at $D_{\text{lat}} = 50$. The purely local collision-stream structure makes LBM the best candidate for GPU parallelisation.

Implementation effort ranges from ∼80 lines for FEM (leveraging NGSolve abstractions, but requiring an external Netgen install) to ∼120 for LBM (Zou–He BCs [8] being the non-trivial part) and ∼150 for FD (half-cell stencils and Poisson boundary treatment).

B. Optimizing the router placement

The optimal router placement consistently lies near the junctions rather than inside a single room. This is to be expected, since junctions provide multiple low-resistance paths into different regions, whereas placing the router within a single room forces signals to cross walls and attenuate. Energy at junctions distributes directly through openings, minimizing wall crossings and producing a more uniform field with higher overall objective values.

The adjoint solver is robust to moderate variations in geometry and wavenumber. However, our optimization focus was on global signal strength, not local maxima. Some rooms, such as Bedroom 1 and the bathroom, receive below-average signal (e.g., 4.9 dB in Bedroom 1), clearly indicating our choice to maximize overall coverage rather than guaranteeing uniform local strength.

Sweeping the wall absorption parameter (imaginary part of the refractive index) showed that the objective score fell sharply as walls become more lossy and eventually flattens out. We also noted a frequency dependency for the optimal router location (k sweep). At the chosen ($k = 1/3$), wall attenuation is already high (≈11dB per wall), making placement crucial for acceptable coverage across all rooms.

VI. Conclusion

For Challenge A, FEM and LBM-MRT both produce the DFG benchmark [10] Strouhal number (St ≈ 0.30) at Re $= 100$. FD fails to sustain shedding. FD survives the longest in stability testing (beyond Re $= 500$) but this is at the cost of physical accuracy. FEM fails at Re $= 200$, and LBM-MRT [7] at Re $= 500$. Overall, FEM is the most accurate near curved boundaries but also the most expensive. LBM has the best potential for parallel and GPU execution. FD provides the simplest implementation with robust stability, though its reliance on volume penalization limits its physical accuracy.

For challenge B, we have identified an optimal router location at $(0.254, 2.890)\,\mathrm{m}$, achieving a total score of $66.218\,\mathrm{dB}$ at an effective frequency of $0.80\,\mathrm{GHz}$ ($\lambda_{\text{eff}} = 37.5\,\mathrm{cm}$). Numerically, the Helmholtz formulation captures diffraction, partial transmission, and reflections, while the chosen wavenumber $k = 1/3$ ensures adequate spatial resolution without excessive computational cost. The adjoint solver efficiently evaluates the entire objective landscape in a single pass for robust determination of global optimal positioning.

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